

ADVANCES IN SPATIAL DATA SCIENCE

Causality and reproducibility

LEVI JOHN WOLF

levi.john.wolf@bristol.ac.uk



LEVI JOHN WOLF (he/him)



Spatializing partisan gerrymandering forensics

PhD Arizona State University, 2017

University of Bristol & Alan Turing Institute (2017-2023)

CONSULTING & ENGAGEMENT

Nextdoor.com

CARTO

MondialRelay

Boundary Commission for England

BCC Living Rent Commission

METHODS

Space-time exploratory data analysis

Spatial optimization

Probabilistic programming

Spatial machine learning

Bayesian spatial econometrics

Local modelling

SUBJECTS

Districting, voting, and elections

Housing and renting in cities

Economic inequality

Segregation, gentrification, and urban change

Species distribution modelling

Urban planning & governance

A little bit about me

THE CITY AND CAUSALITY

MUCH ADO ABOUT NULL THINGS

Rethinking causality in city science & spatial analysis

THE CITY AND CAUSALITY

what's replication got to do with it?

MUCH ADO ABOUT
NULL THINGS

Rethinking causality in city science & spatial analysis

Environment and Planning B: Urban Analytics and City Science

Impact Factor: 3.5

5-Year Impact Factor: 3.9

Editorial board

Managing Editor

Michael Batty

University College London, UK

Editors

Seraphim Alvanides

Northumbria University, UK

Daniel Arribas-Bel

University of Liverpool, UK

Andrew Crooks

State University of New York at Buffalo

Linda See

International Institute for Applied Systems Analysis (IIASA), Austria

Levi Wolf

University of Bristol, UK

Cities, laws, and history

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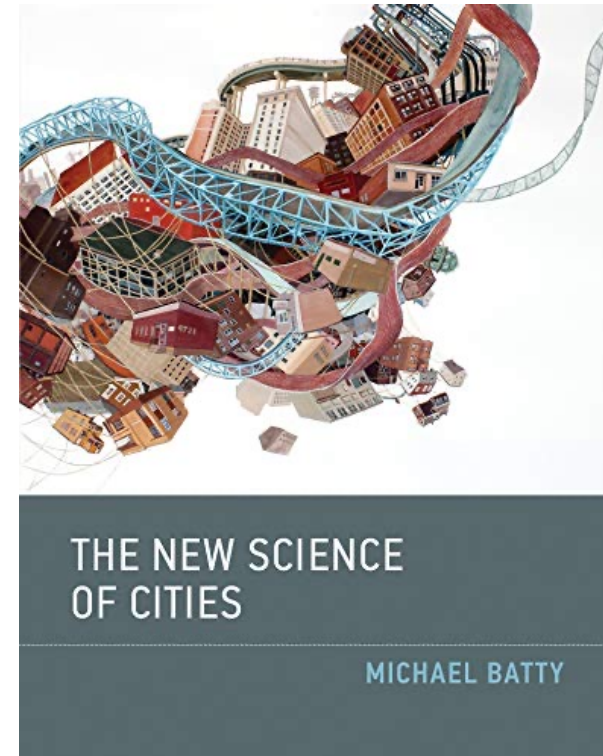
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THE NEW S OF CITIES

Luis M. A. Bettencourt

INTRODUCTION TO Urban Science

Evidence and Theory
of Cities as
Complex Systems

Cities, laws, and history

A roundtable discussion: Defining urban data science

EPB: Urban Analytics and City Science

2019, Vol. 46(9) 1756–1768

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DOI: 10.1177/2399808319882826

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Organizers

Wei Kang, Taylor Oshan, Levi John Wolf

Participants

Geoff Boeing, Vanessa Frias-Martinez,
Song Gao, Ate Poorthuis, Wenfei Xu

*Urban Data Science is sometimes **the dilettante practice** of doing things with software and data about living in cities [...] to be a science, urban data science both has to **build from theory** and **give back to theory**.*





A Research Agenda for

Spatial Analysis

Edited by
Levi John Wolf
Richard Harris
Alison Heppenstall

ELGAR
RESEARCH
AGENDAS



A challenge for spatial analysis, as for many areas of applied social science, data science and statistics, is **answering the question of causality ... what caused something to happen or arise where and when it did?**

(Wolf, Harris, Heppenstall)

generative modelling done right can both contribute to a new form of causal inference and to the larger program of social sciences: the simultaneous search for generalised explanations of social phenomena and recognition of the uniqueness of historical events.
(Cottineau)

there are causal processes operating at a level (or levels) beyond the individual, but if the level included in an analysis does not correspond with the level(s) of the process then it is possible it will be mis-estimated
(Petrovic, Manley, & Van Ham)

Beyond open science: Data, code, and causality

Levi John Wolf

EPB: Urban Analytics and City Science

2023, Vol. 50(9) 2333–2336

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Indeed, city science is relatively unusual in the social sciences, in that one can still often find work seeking ‘laws’



A COMPUTER MOVIE SIMULATING URBAN GROWTH IN THE DETROIT REGION

W. R. TOBLER

University of Michigan

The LAWS of MIGRATION.

By E. G. RAVENSTEIN, Esq., F.R.G.S.

HUMAN BEHAVIOR

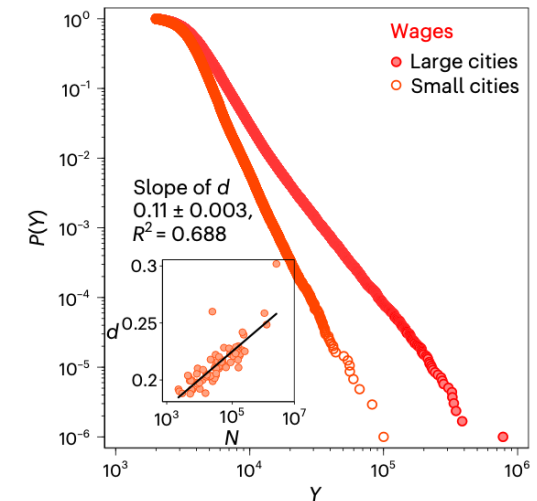
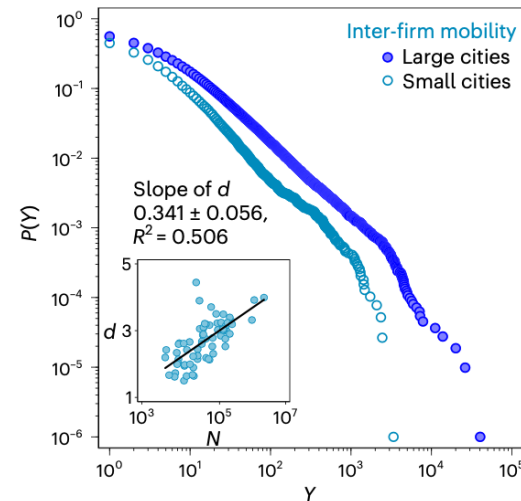
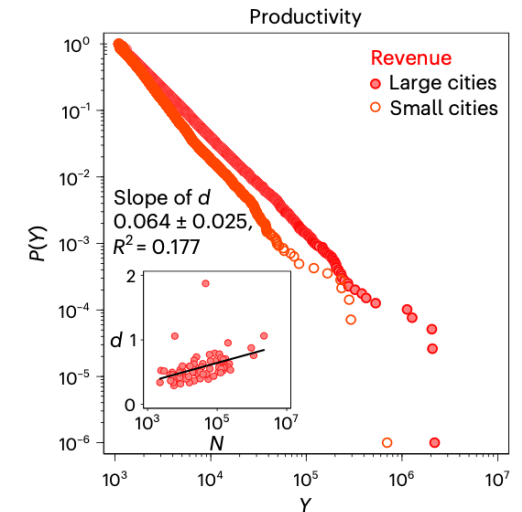
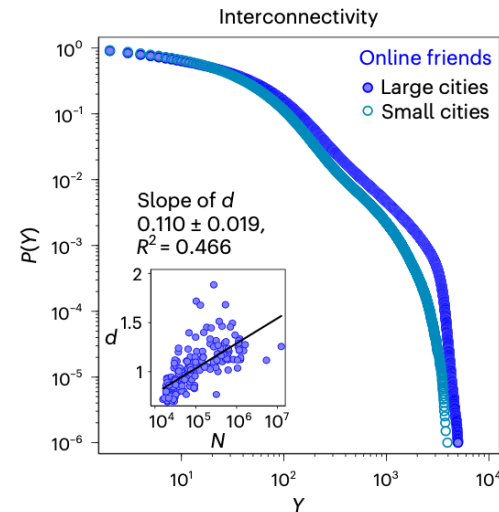
AND

THE PRINCIPLE OF LEAST EFFORT



An Introduction to Human Ecology

GEORGE KINGSLEY ZIPF, Ph.D.



Arvidsson et al. (2023)

laws

Replication across space and time must be weak in the social and environmental sciences

Michael F Goodchild & Wenwen Li

Proceedings of the National Academy of Sciences, 2021

doi.org/10.1073/pnas.2015759118

GEOGRAPHIC INFORMATION SCIENCE II: Mesogeography: social physics, GIScience, and the quest for geographic knowledge

Harvey J. Miller

Progress in Human Geography, 2018

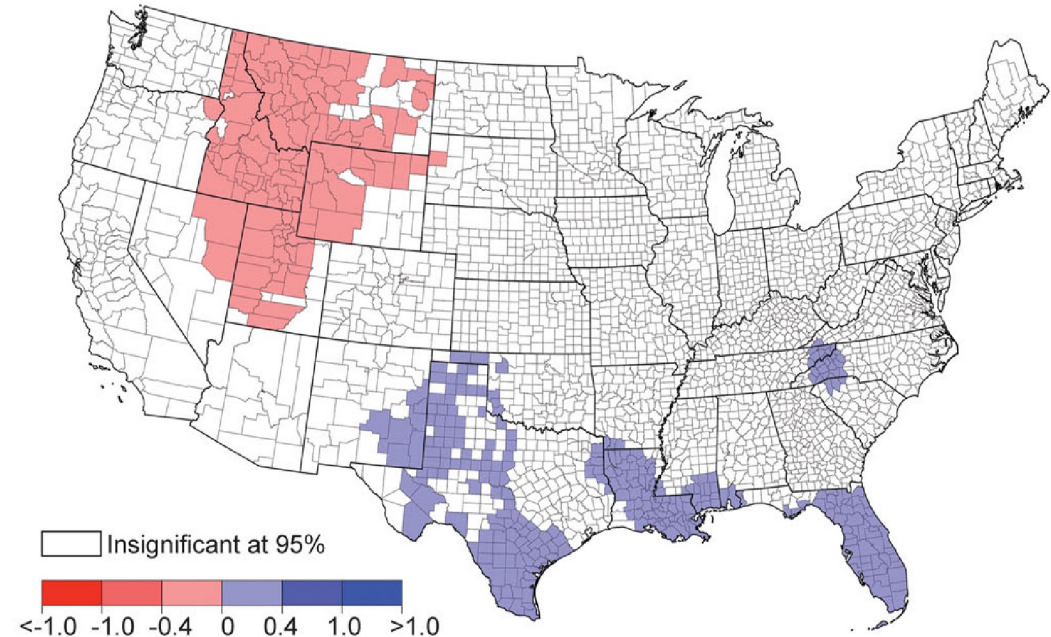
doi.org/10.1177/0309132517712154

(B)

MGWR Local Parameter Estimates of Turnout

Bandwidth: 117

(Global OLS Est.: 0.168*)



Scale, Context, and Heterogeneity: A Spatial Analytical Perspective on the 2016 U.S. Presidential Election

A. Stewart Fotheringham,^{*} Ziqi Li,[†]  and Levi John Wolf[‡] 

Contextual 'laws'?

Rethinking Causality in Quantitative Human Geography

Mirah Zhang

Levi John Wolf

Rather than accepting Goodchild & Li's "weak replicability," we must rigorously work together to understand where and why replicability fails. Spatial analysts must get specific about what "scale" or "context" does

preprint



Beyond open science: Data, code, and causality

Levi John Wolf

EPB: Urban Analytics and City Science

2023, Vol. 50(9) 2333–2336

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*Many contemporary definitions of causality,
such as Lewis’s “counterfactual theory of
causation,” will result in very different
replication practices*





Urban scaling laws arise from within-city inequalities

Received: 12 August 2021

Martin Arvidsson¹✉, Niclas Lovsjö¹✉ & Marc Keuschnigg^{1,2}✉

Accepted: 6 December 2022

The causal processes underlying [inequality within cities] cities [is] an indispensable element of urban scaling

Law-as-cause

fails if inequality does not drive scaling swh.

*“Weak Replicability”
(Goodchild & Li 2021)*

still holds if inequality drives scaling differently (or not at all) in different places/times

*“Strong contextuality”
(Zhang & Wolf, 2024)*

holds iff we can specify where/when inequality does not drive scaling



Beyond open science: Data, code, and causality

Levi John Wolf

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*To really achieve the open science aim of
“replication,” we need to have theories
that can be replicated.*



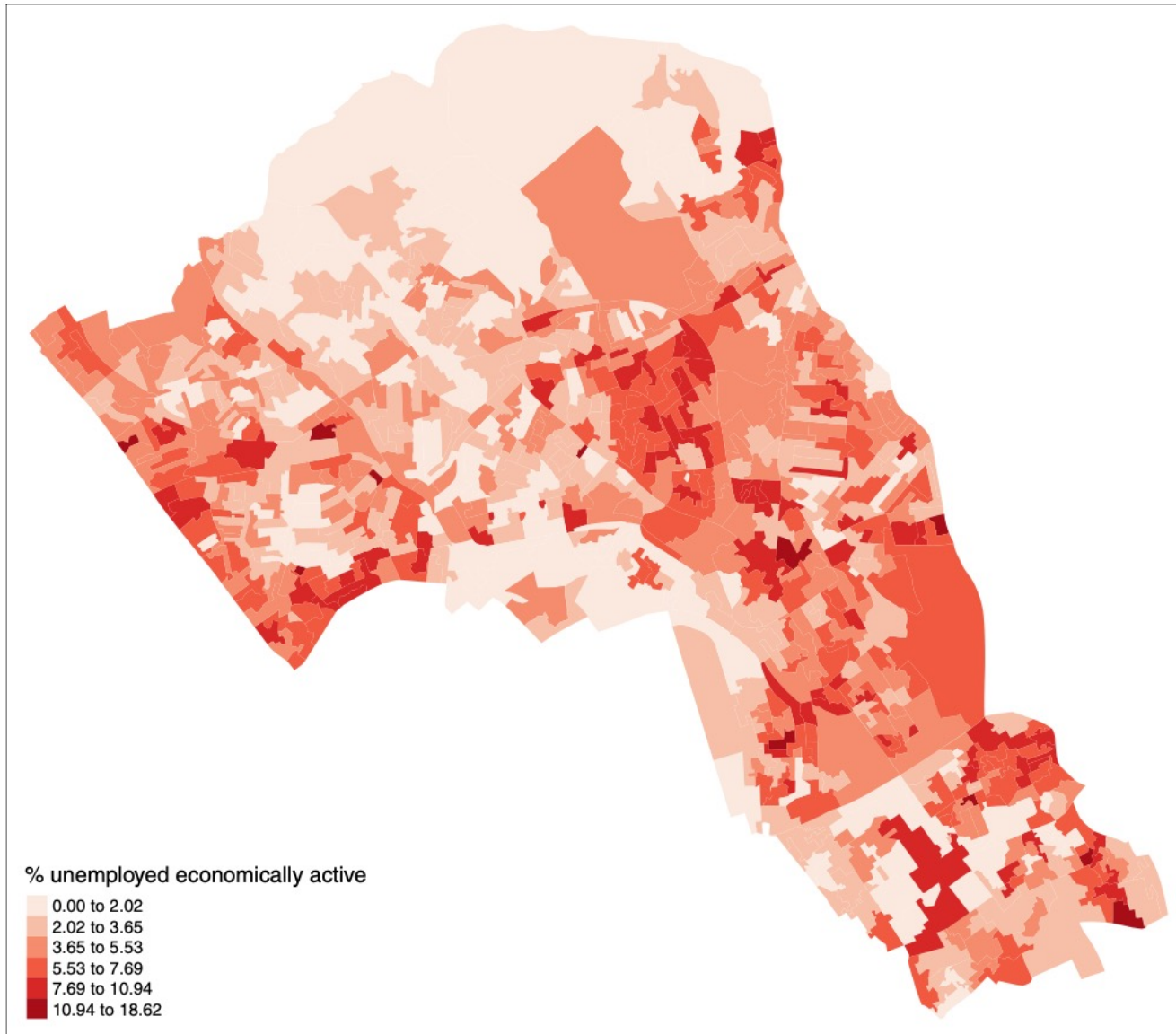
THE CITY AND CAUSALITY

what's replication got to do with it?

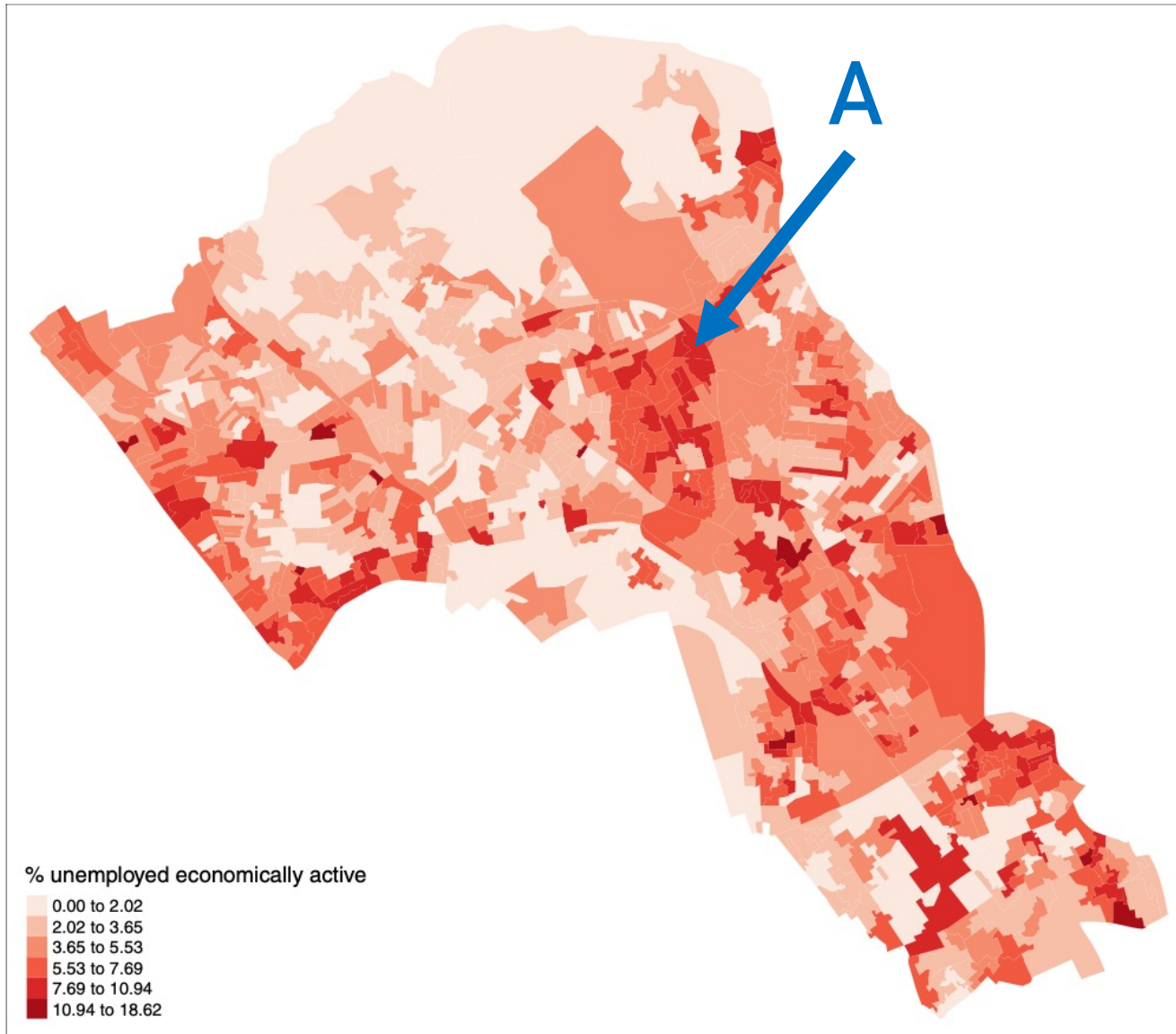
MUCH ADO ABOUT NULL THINGS

a recent replication challenge

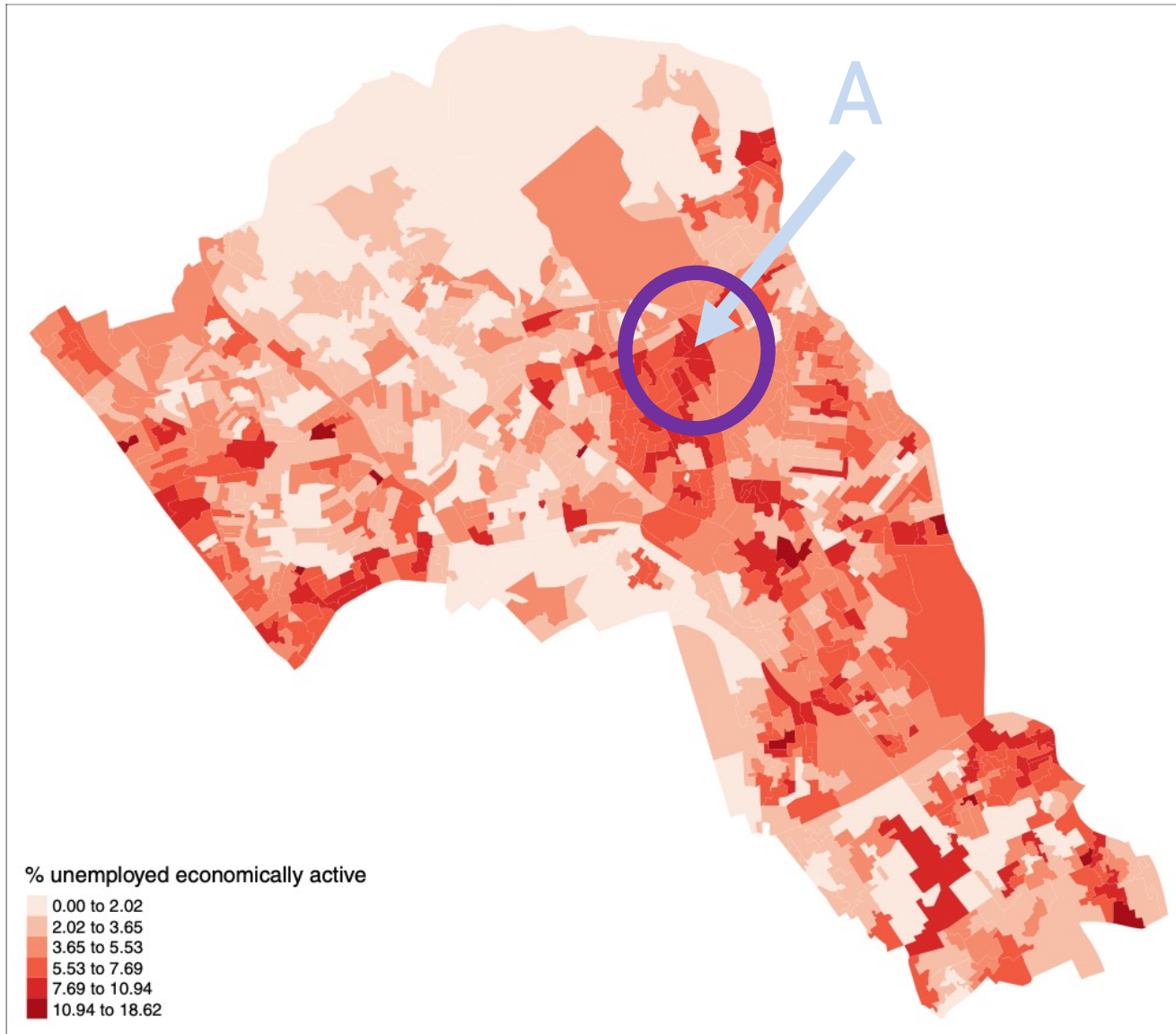
Rethinking causality in city science & spatial analysis



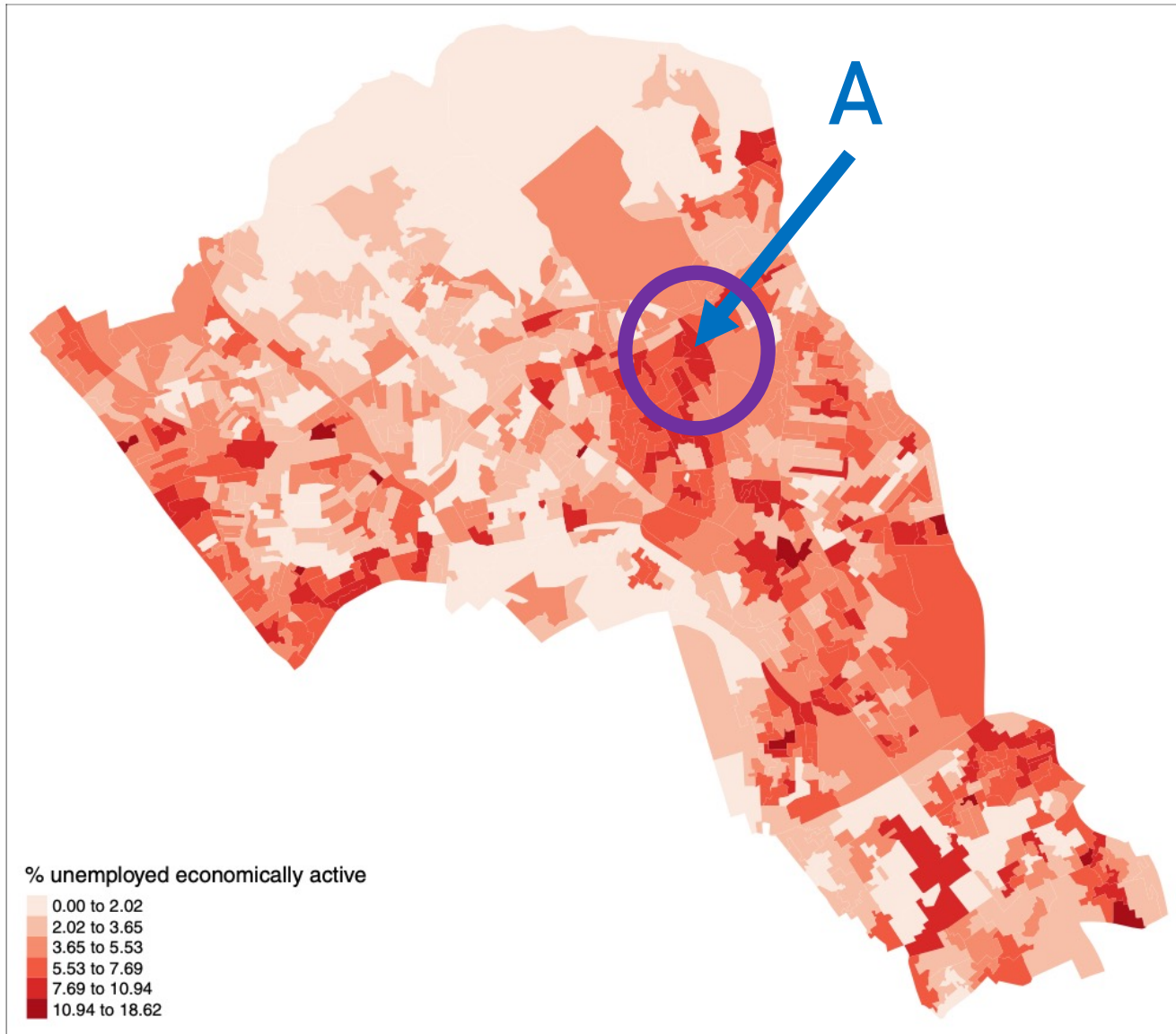
Local statistics: find outliers and/or clusters



Local statistics: find outliers and/or clusters

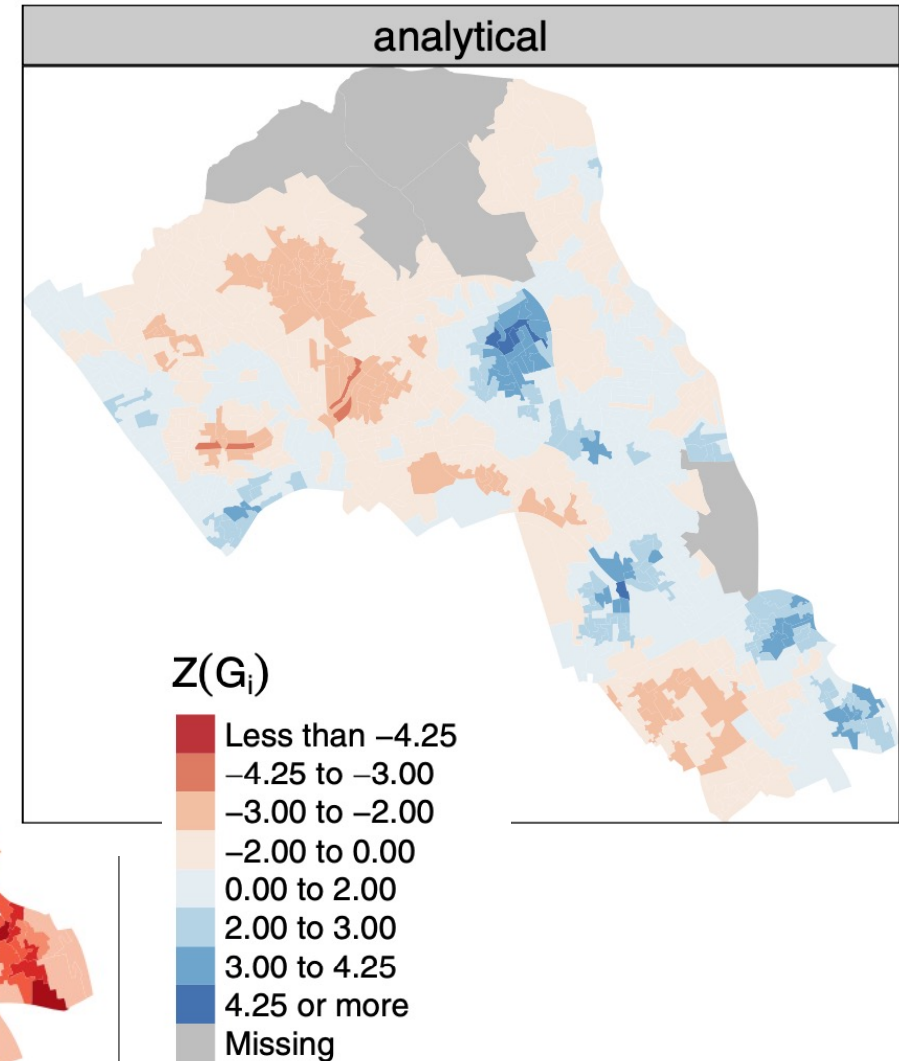
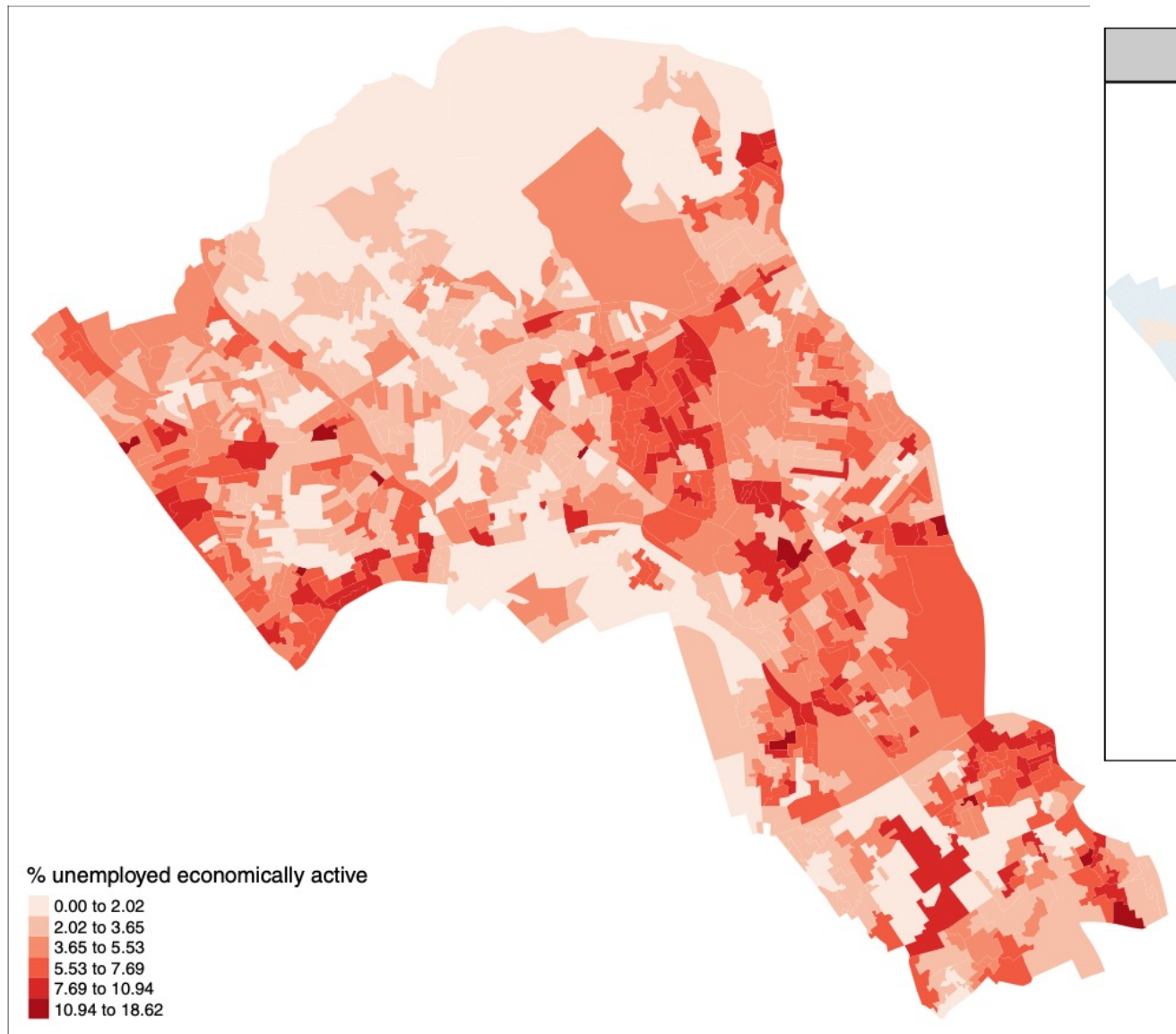


Local statistics: find outliers and/or clusters

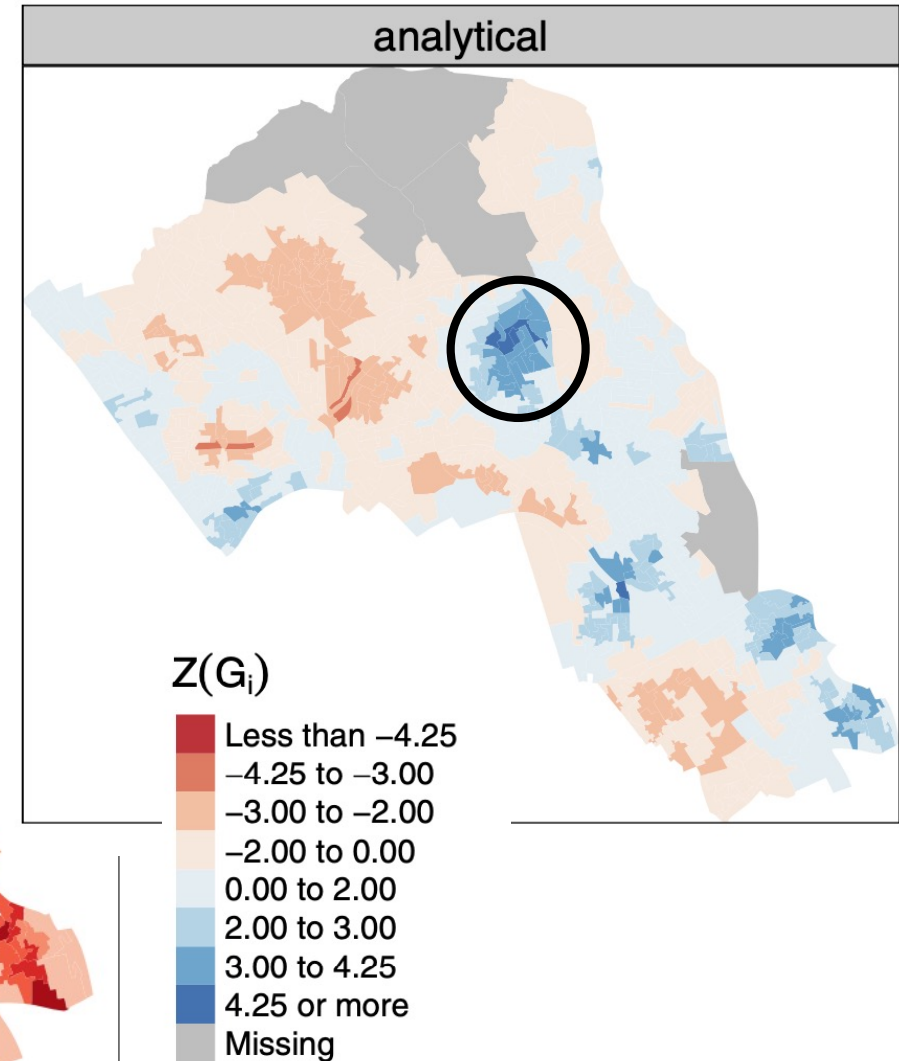
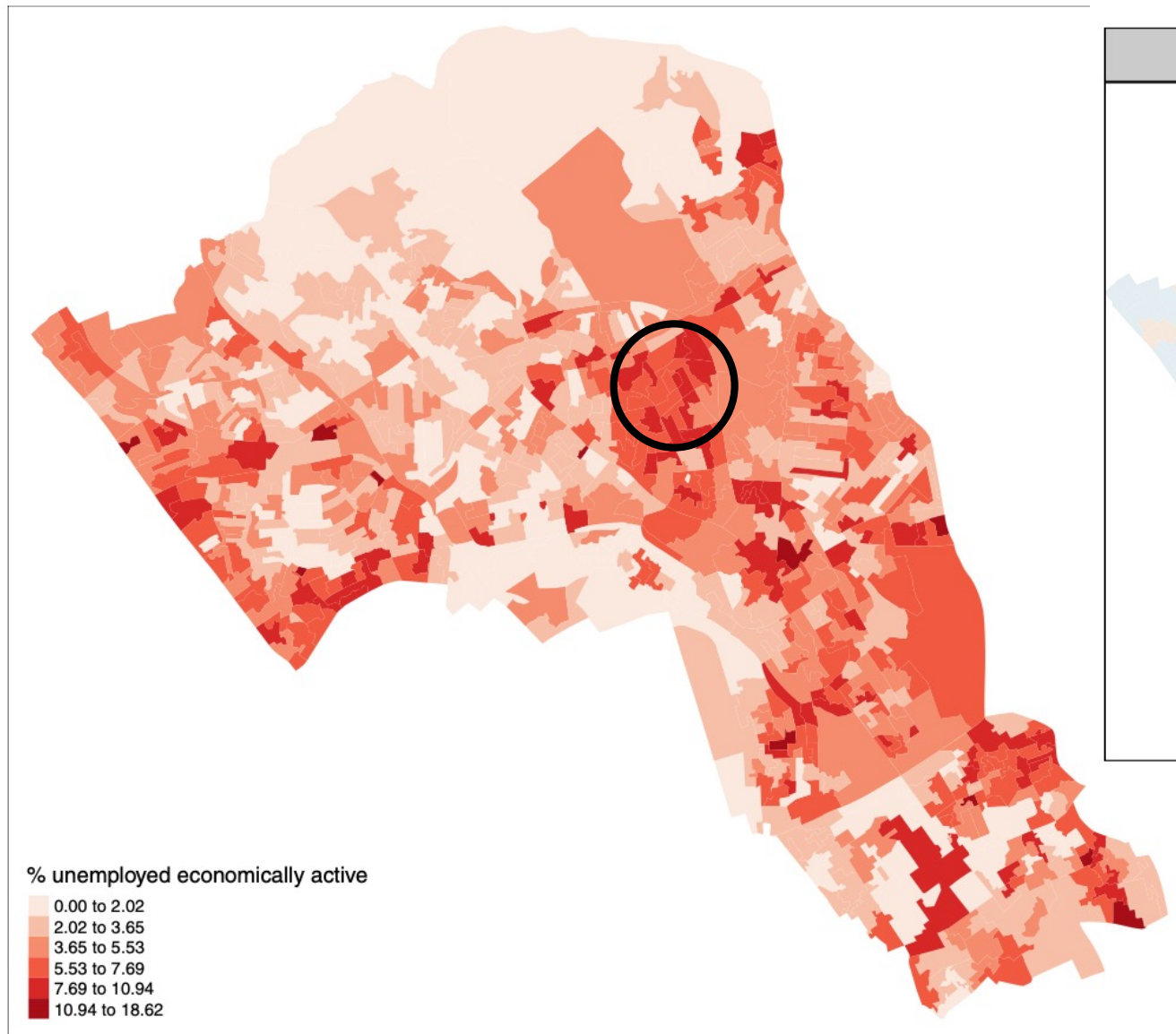


Is this site
unusual for the
values in its
surrounding
area?

Local statistics: find outliers and/or clusters



Local statistics: find outliers and/or clusters



Local statistics: find outliers and/or clusters

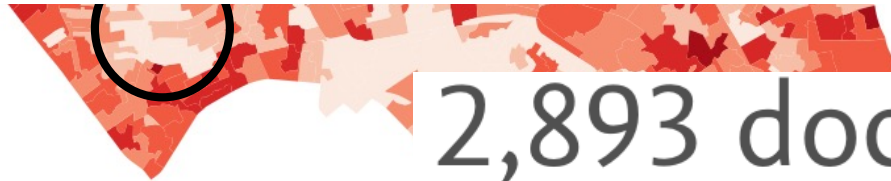
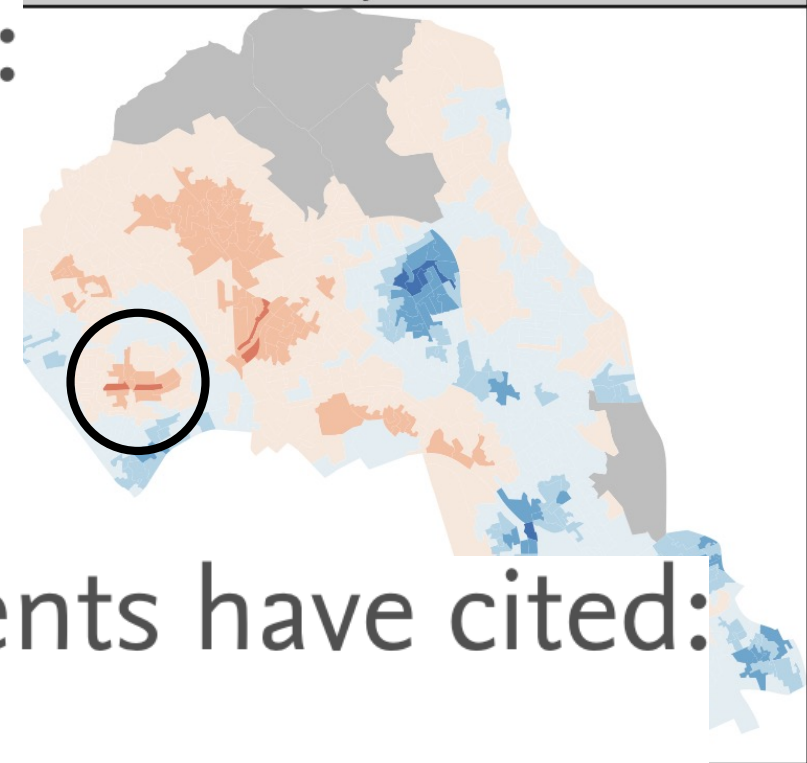
5,679 documents have cited:

Local Indicators of Spatial Association—LISA

Anselin L.

(1995) Geographical Analysis, 27 (2) , pp. 93-115.

analytical



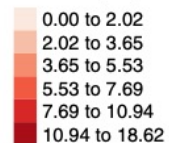
2,893 documents have cited:

The Analysis of Spatial Association by Use of Distance Statistics

Getis A., Ord J.K.

(1992) Geographical Analysis, 24 (3) , pp. 189-206.

% unemployed economically active



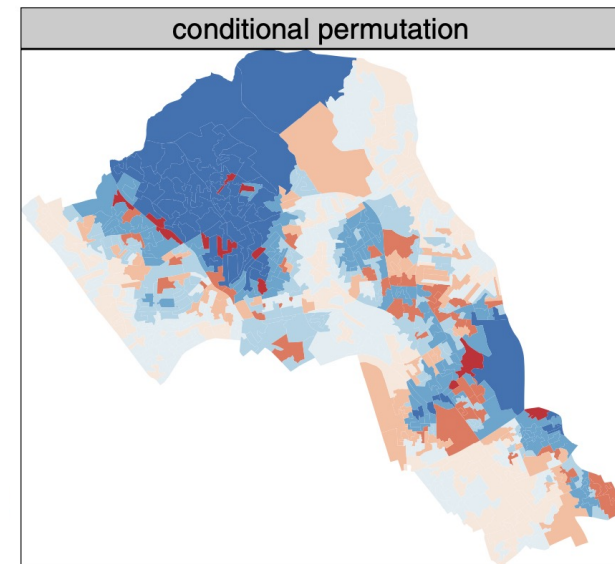
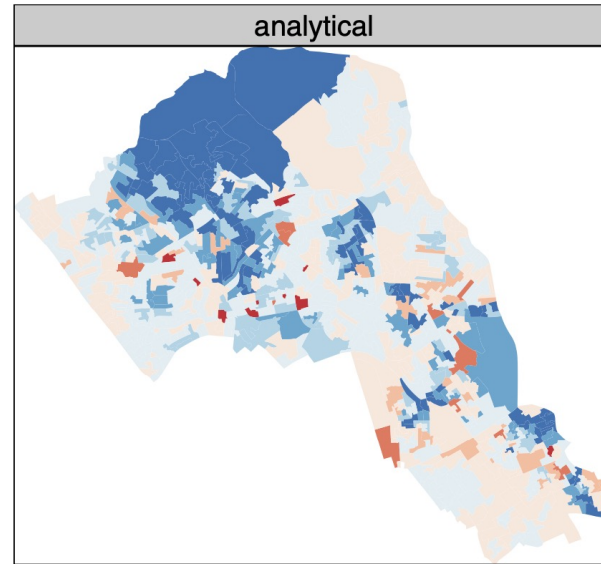
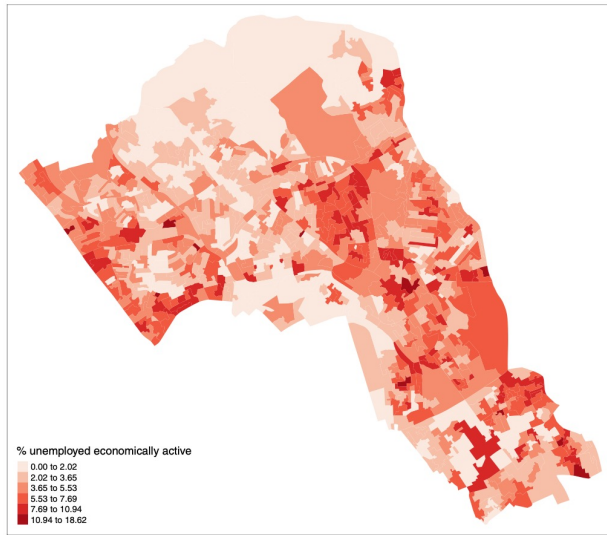
Local statistics: find outliers and/or clusters

Comparing implementations of global and local indicators of spatial association

Roger S. Bivand¹  • David W. S. Wong²

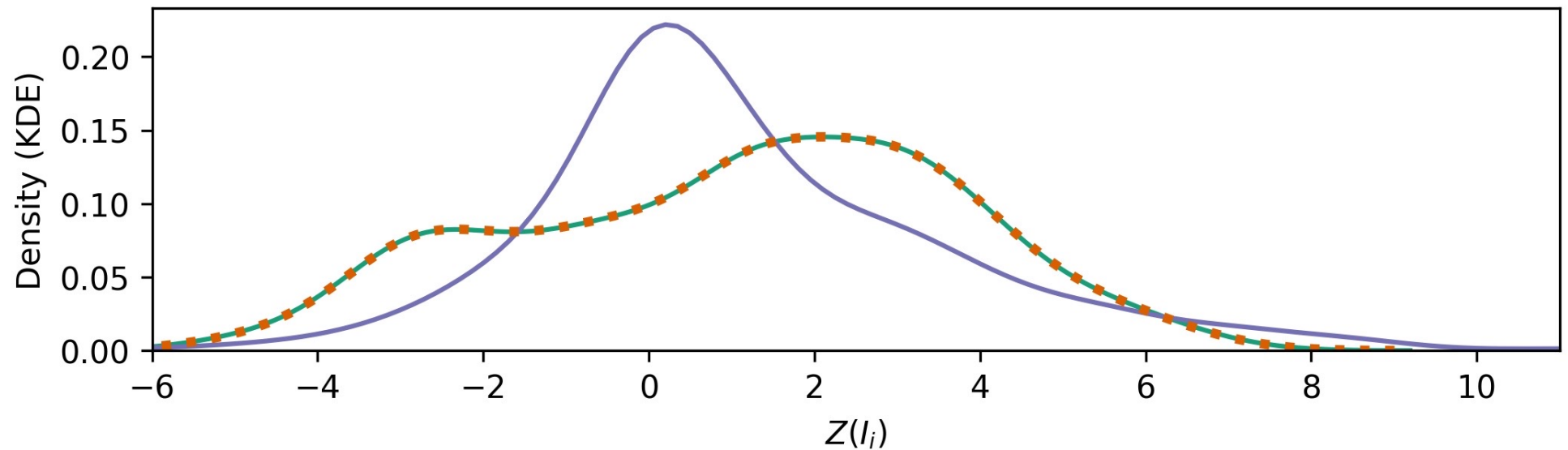
User choices for local measures both of software and of inferential method over and above the handling of multiple comparisons will have consequences for conclusions



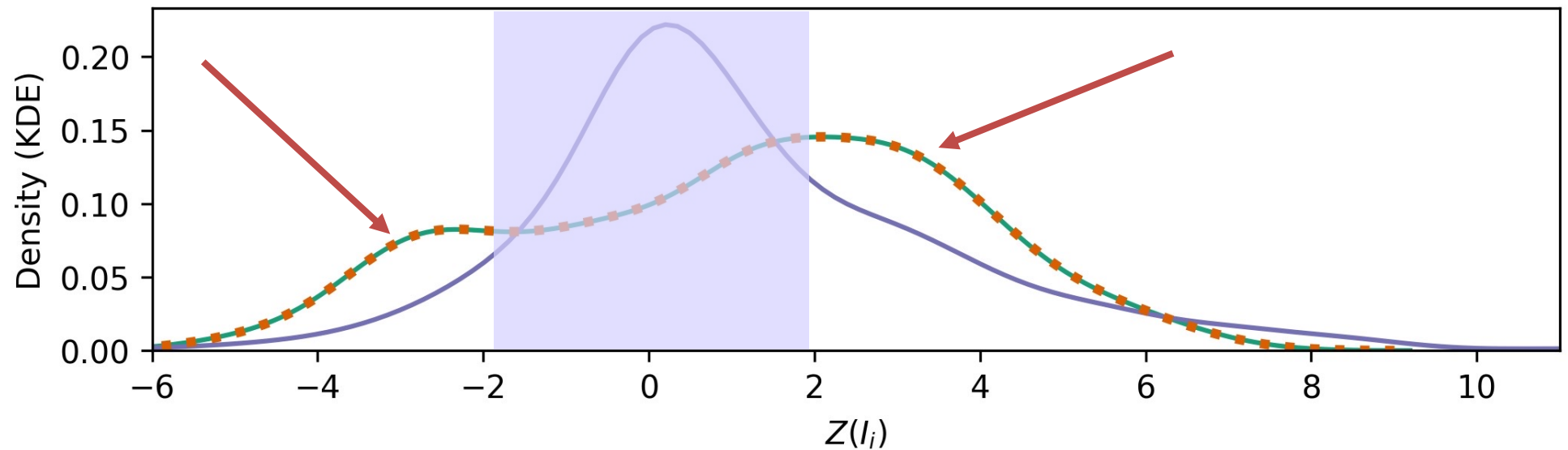


[T]he spatial patterns of Z values generated by conditional permutation for local measures differ considerably from those calculated using analytical methods.





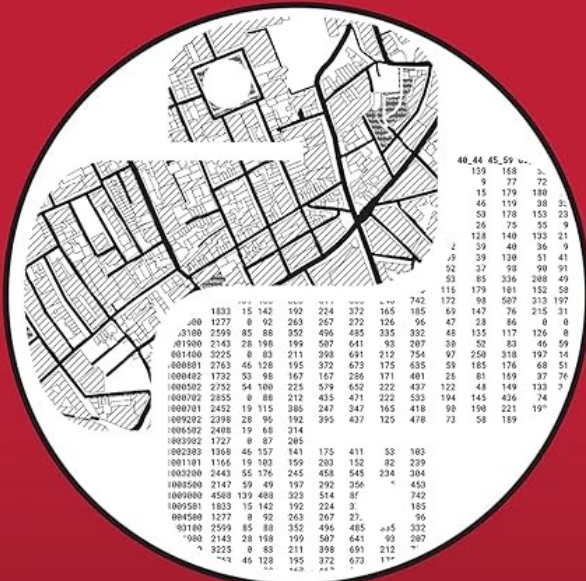
The permutation-based local z-scores were much more likely to be statistically significant [...] Analytical inference yielded 267 significant OAs whereas permutation-based inference yielded 429. [...] OAs were statistically significant using analytical inference but not using permutation-based inference, and vice versa.



The permutation-based local z-scores were much more likely to be statistically significant [...] Analytical inference yielded 267 significant OAs whereas permutation-based inference yielded 429. [...] OAs were statistically significant using analytical inference but not using permutation-based inference, and vice versa.

Texts in Statistical Science

Geographic Data Science with Python



Sergio Rey
Dani Arribas-Bel
Levi John Wolf

 **CRC Press**
Taylor & Francis Group

A CHAPMAN & HALL BOOK

The R Series

Spatial Data Science With Applications in R



Edzer Pebesma
Roger Bivand

 **CRC Press**
Taylor & Francis Group
A CHAPMAN & HALL BOOK



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rs

differ



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The Importance of Null Hypotheses: Understanding Differences in Local Moran's I_i under Heteroskedasticity

Jeffery Sauer¹ , Taylor Oshan¹, Sergio Rey², Levi John Wolf³

We see two different ways the divergence ... could come about. Bivand & Wong (2018)'s theory is that local heteroskedasticity in the data is causing their ... estimates to diverge. Our theory is that different null hypotheses drive the observed differences.



ANALYTIC

SYNTHETIC

ANALYTIC

Assume normal observations
distributed over a regular grid

Derive the expectation $E[\Gamma]$

Use to define $\text{Var}[\Gamma]$

Construct z-score

Assess against a standard normal
distribution: $|z_i| > 1.96$

Hope it generalizes!

SYNTHETIC

ANALYTIC

SYNTHETIC

$$\mathbf{E}[I_i] = - \sum_{j \neq i} \frac{w_{ij}}{n-1}$$

$$\begin{aligned} \mathbf{Var}[I_i] = & \frac{w_{i(2)}(n-b_2)}{n-1} \\ & + \frac{2w_{i(kh)}2b_2-n}{(n-1)(n-2)} \\ & + \left[\sum_{i \neq j} \frac{w_{ij}}{n-1} \right]^2 \end{aligned}$$

ANALYTIC

$$\mathbf{E}[I_i] = - \sum_{j \neq i} \frac{w_{ij}}{n-1}$$

$$\begin{aligned} \mathbf{Var}[I_i] = & \frac{w_{i(2)}(n-b_2)}{n-1} \\ & + \frac{2w_{i(kh)}2b_2-n}{(n-1)(n-2)} \\ & + \left[\sum_{i \neq j} \frac{w_{ij}}{n-1} \right]^2 \end{aligned}$$

SYNTHETIC

Assume the data that you *do have* is like data you *could have*

Hold the value at the site you're testing constant

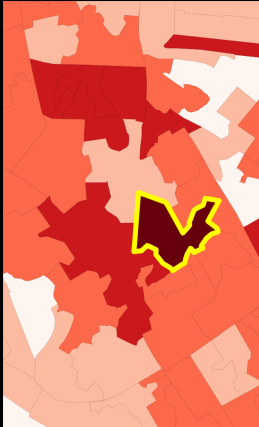
Shuffle the *rest* of the map

Compute a new local statistic

Repeat many times to obtain distribution of replicates,

Describe your site *relative to* the replicates

REAL



SYNTHETIC

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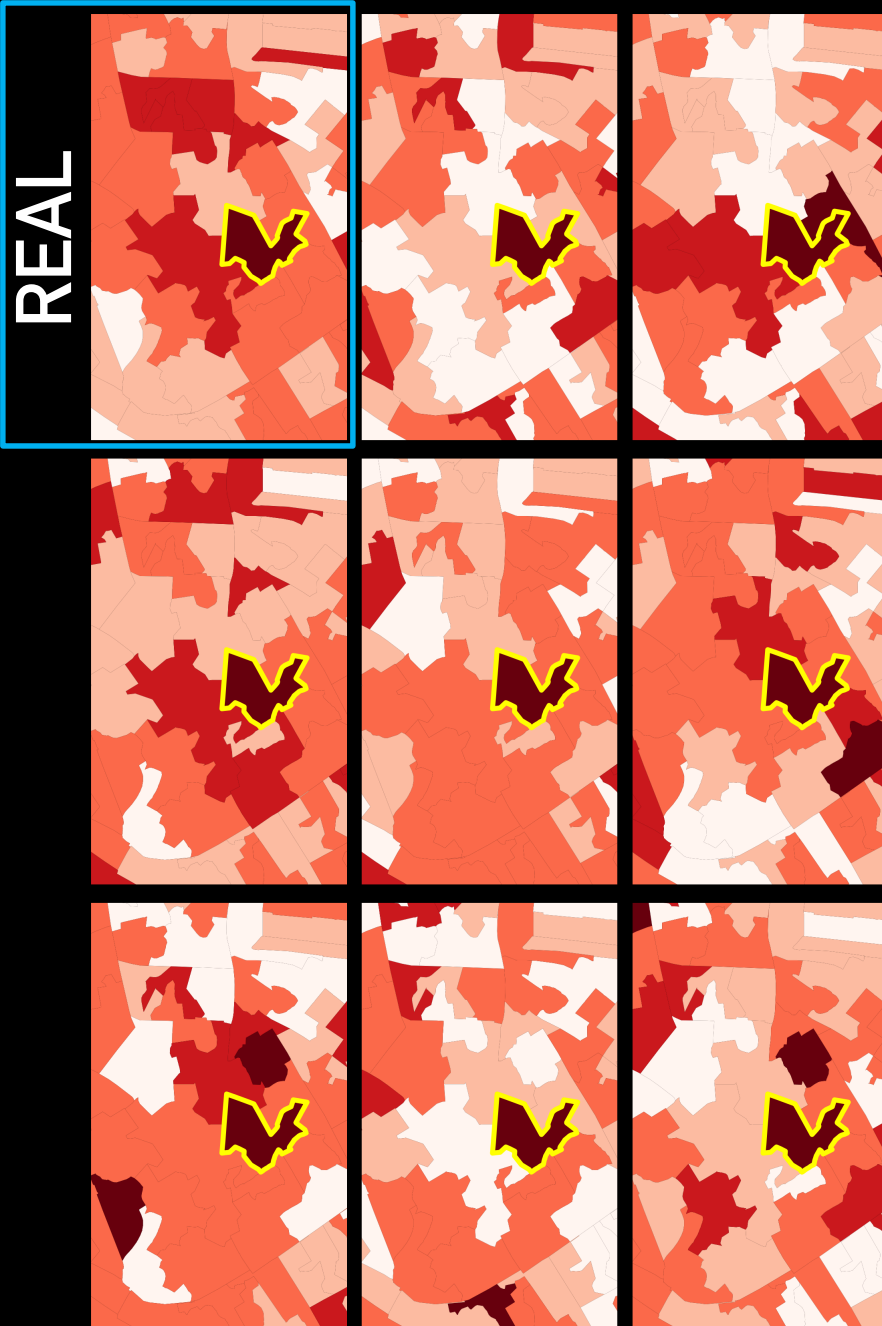
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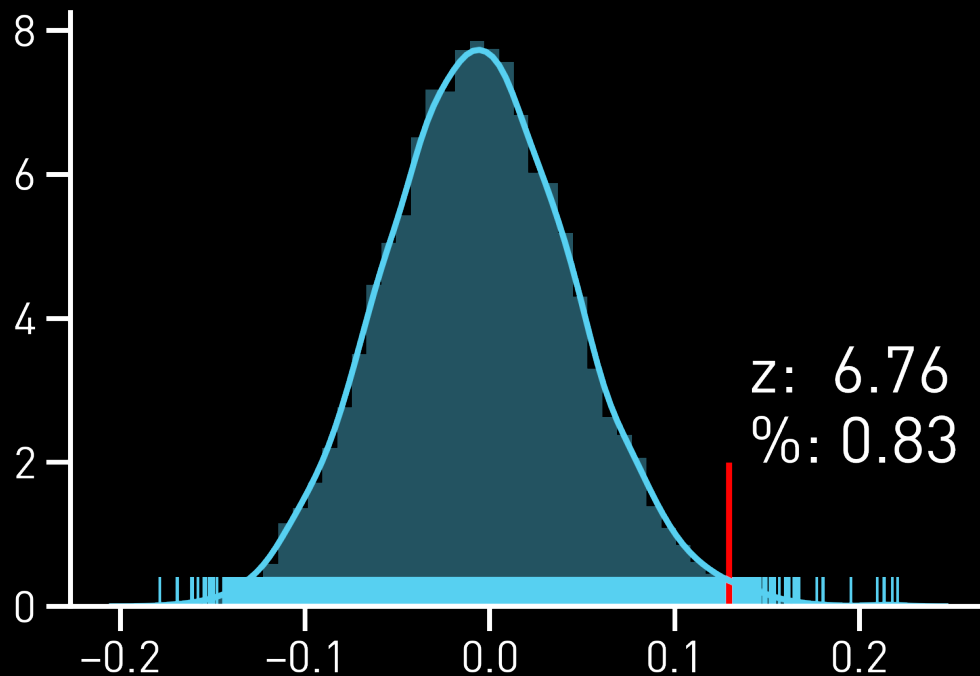
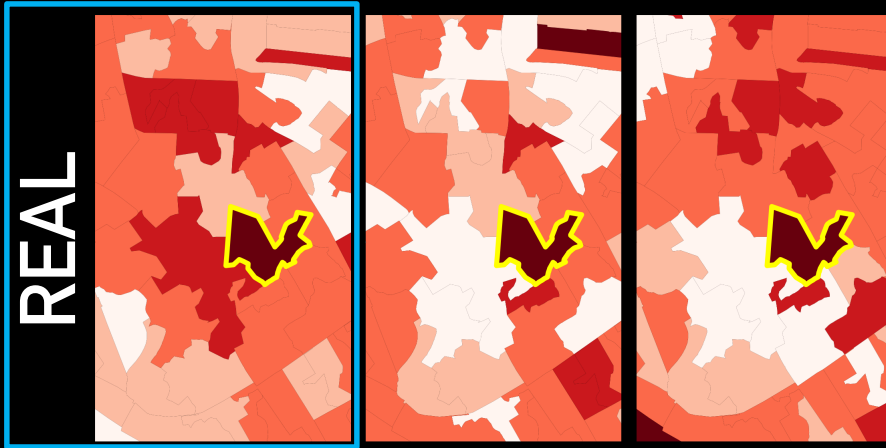
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ANALYTIC

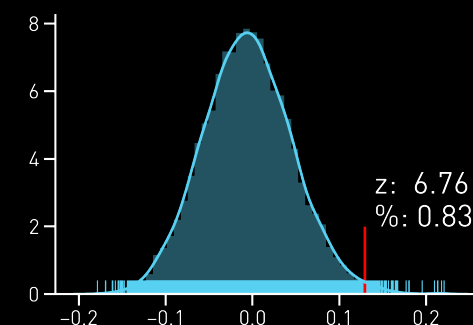
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$$\begin{aligned} \mathbf{Var}[I_i] = & \frac{w_{i(2)}(n-b_2)}{n-1} \\ & + \frac{2w_{i(kh)}2b_2-n}{(n-1)(n-2)} \\ & + \left[\sum_{i \neq j} \frac{w_{ij}}{n-1} \right]^2 \end{aligned}$$

SYNTHETIC



	"Total" Randomization Null <i>shuffle all values, so each value is equally likely at any site</i>	"Conditional" Randomization Null <i>for each site, the site's value is fixed and remaining sites are shuffled</i>
Analytical estimator <i>closed-form mathematical expressions for test statistics</i>	Originally in Anselin (1995). Implemented in pysal and spdep . Used in Bivand & Wong (2018). Denoted here as $E[I_i]$ and $Var[I_i]$	Originally in Sokal (1998). Not implemented in pysal and spdep or considered by Bivand & Wong (2018). Denoted here as $E_c[I_i]$ and $Var_c[I_i]$
Permutation estimator <i>test statistics computed from set of simulated maps</i>	Not considered or implemented.	Originally in Anselin (1995). Implemented in pysal . Used in Bivand and Wong (2018). Denoted here as $E_p[I_i]$ and $Var_p[I_i]$



	"Total" Randomization Null <i>shuffle all values, so each value is equally likely at any site</i>	"Conditional" Randomization Null <i>for each site, the site's value is fixed and remaining sites are shuffled</i>
Analytical estimator <i>closed-form mathematical expressions for test statistics</i>	Originally in Anselin (1995). Implemented in pysal and spdep . Used in Bivand & Wong (2018). Denoted here as $E[I_i]$ and $Var[I_i]$	Originally in Sokal (1998). Not implemented in pysal and spdep or considered by Bivand & Wong (2018). Denoted here as $E_c[I_i]$ and $Var_c[I_i]$
Permutation estimator <i>test statistics computed from set of simulated maps</i>	Not considered or implemented.	Originally in Anselin (1995). Implemented in pysal . Used in Bivand and Wong (2018). Denoted here as $E_p[I_i]$ and $Var_p[I_i]$

$$\mathbf{E}[I_i] = - \sum_{j \neq i} \frac{w_{ij}}{n-1}$$

$$\mathbf{Var}[I_i] = \frac{w_{i(2)}(n-b_2)}{n-1} + \frac{2w_{i(kh)}2b_2-n}{(n-1)(n-2)} + \left[\sum_{i \neq j} \frac{w_{ij}}{n-1} \right]^2$$

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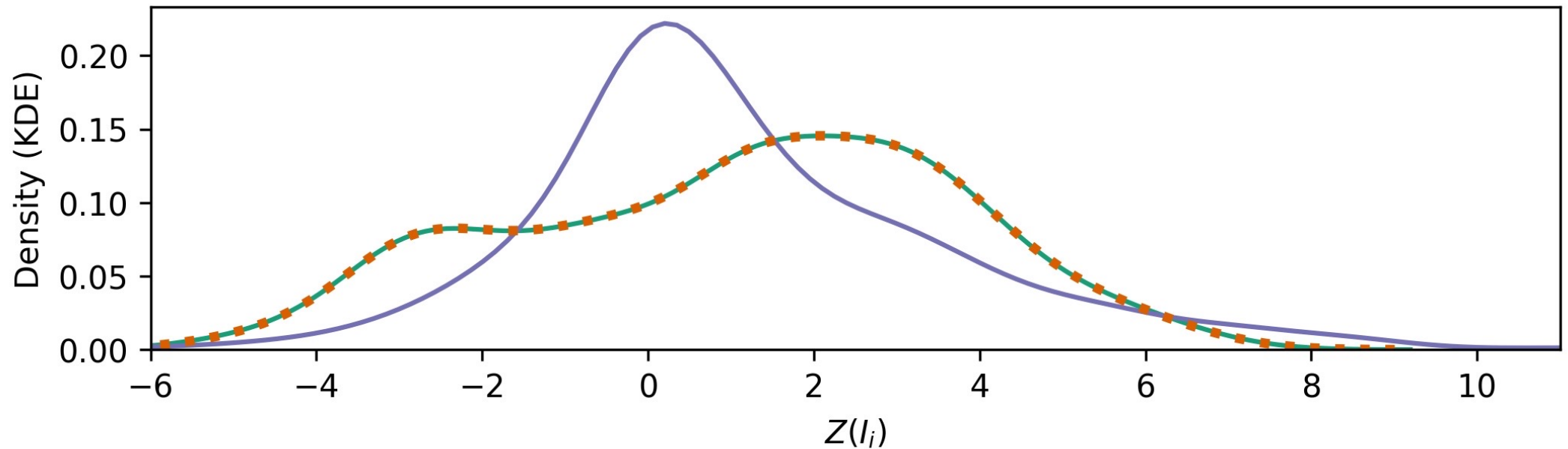
$$\mathbf{E}_c[I_i] = -z_i^2 \sum_{i \neq j} \frac{w_{ij}}{n-1}$$

$$\mathbf{Var}_c[I_i] = \left[\frac{z_i}{m_2} \right]^2 \left[\frac{n}{n-2} \right] \left[w_{i(2)} - \frac{\left(\sum_{i \neq j} w_{ij} \right)^2}{n-1} \right] \left[m_2 - \frac{z_i^2}{n-1} \right]$$

	"Total" Randomization Null <i>shuffle all values, so each value is equally likely at any site</i>	"Conditional" Randomization Null <i>for each site, the site's value is fixed and remaining sites are shuffled</i>
Analytical estimator <i>closed-form mathematical expressions for test statistics</i>	Originally in Anselin (1995). Implemented in pysal and spdep . Used in Bivand & Wong (2018). Denoted here as $E[I_i]$ and $Var[I_i]$	Originally in Sokal (1998). Not implemented in pysal and spdep or considered by Bivand & Wong (2018). Denoted here as $E_c[I_i]$ and $Var_c[I_i]$
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$$\mathbf{E}[I_i] = - \sum_{j \neq i} \frac{w_{ij}}{n-1} \quad \mathbf{E}_c[I_i] = - \boxed{z_i^2} \sum_{i \neq j} \frac{w_{ij}}{n-1}$$

Null likelihood for the stat depends on the observed value at the site!



	$Z(I_i)$	$Z_p(I_i)$	$Z_c(I_i)$
$Z(I_i)$	1.00	0.85	0.85
$Z_p(I_i)$	0.85	1.00	1.00
$Z_c(I_i)$	0.85	1.00	1.00

- Empirical "conditional randomization" $Z_p(I_i)$
- Analytical "total randomization" $Z(I_i)$
- - - Analytical "conditional randomization" $Z_c(I_i)$



Same null, same findings over different estimators

The Importance of Null Hypotheses: Understanding Differences in Local Moran's I_i under Heteroskedasticity



Jeffery Sauer¹ , Taylor Oshan¹, Sergio Rey², Levi John Wolf³

A vast body of work has expanded upon Anselin (1995). But, we [still] tend to distinguish the "total" and "conditional" null by computation, not conceptualization. Our null hypotheses should be more explicitly acknowledged, however, because they are themselves design decisions.

We do not have the right balance

- a. Expecting something to replicate &
- b. Post hoc justifying with spatial or scale “complexity” when it doesn’t

Replication across space and time must be weak in the social and environmental sciences

Michael F Goodchild & Wenwen Li

Proceedings of the National Academy of Sciences, 2021

doi.org/10.1073/pnas.2015759118

**GEOGRAPHIC INFORMATION SCIENCE II:
Mesogeography: social physics, GIScience,
and the quest for geographic knowledge**

Harvey J. Miller

Progress in Human Geography, 2018

doi.org/10.1177/0309132517712154

“Democratizing” methods
makes our defaults “political!”

- Some questions were never answered, but stats is a science of defaults
- Empirical research now scales well, but theoretical understanding is contextual!

**TESTING FOR LOCAL SPATIAL
AUTOCORRELATION IN THE
PRESENCE OF GLOBAL
AUTOCORRELATION**

J. Keith Ord & Arthur Getis

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accepted,
preprint

Confounded Local Inference: Extending Local Moran Stats to Handle Confounding

Levi John Wolf



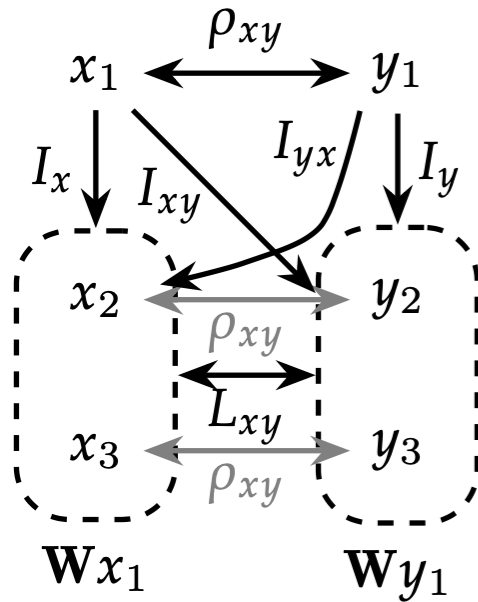
Connecting theory to (local) statistics



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Classic Statistics

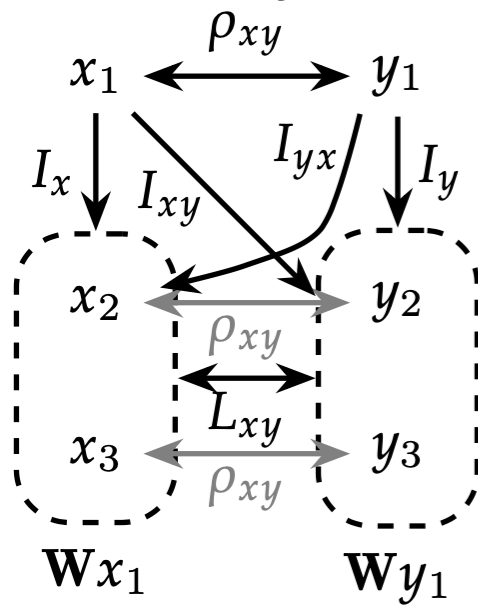
Connecting theory to (local) statistics



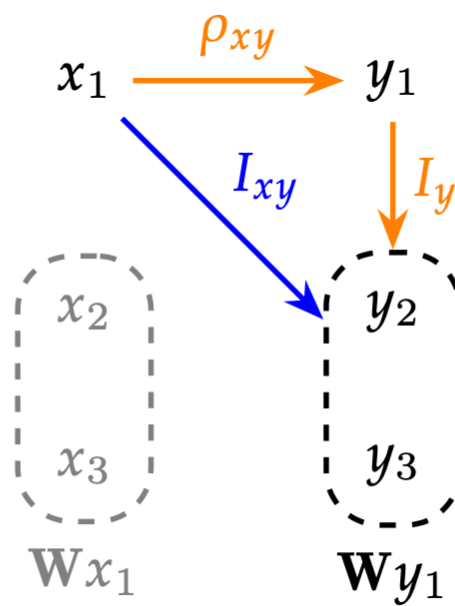
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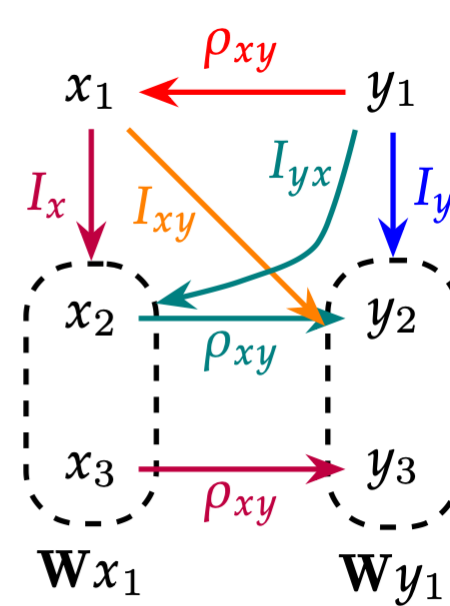
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Classic Statistics



Partial $I_{x|y}$



Full $I_{x \rightarrow y}$

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Name		Statistic
Univariate	$\mathbf{I}_y$	$y \circ \mathbf{W}y$
Bivariate	$\mathbf{I}_{xy}$	$x \circ \mathbf{W}y$
Partial	$\mathbf{I}_{y x}$	$\mathbf{I}_y - \rho_{xy}\mathbf{I}_{xy}$
Auxiliary	$\mathbf{I}_{x \rightarrow y}$	$\mathbf{I}_{y x} - \rho_{xy}\mathbf{I}_{yx} - \rho_{xy}^2\mathbf{I}_x$

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Name		Statistic	
Univariate	$\mathbf{I}_y$	$y \circ \mathbf{W}y$	<i>Are house prices high here?</i>
Bivariate	$\mathbf{I}_{xy}$	$x \circ \mathbf{W}y$	<i>Are big houses expensive here?</i>
Partial	$\mathbf{I}_{y x}$	$\mathbf{I}_y - \rho_{xy}\mathbf{I}_{xy}$	<i>Is this house's price unusual for this area, given its size?</i>
Auxiliary	$\mathbf{I}_{x \rightarrow y}$	$\mathbf{I}_{y x} - \rho_{xy}\mathbf{I}_{yx} - \rho_{xy}^2\mathbf{I}_x$	<i>Is the "extra" you pay for this house given its size unusual for this area?</i>

Connecting theory to (local) statistics

THE CITY AND CAUSALITY

*replication is needed
for city science to work*

MUCH ADO ABOUT
NULL THINGS

*and replication is
harder than it looks!*

Rethinking causality in city science & spatial analysis

ADVANCES IN SPATIAL DATA SCIENCE

Causality and reproducibility

LEVI JOHN WOLF

levi.john.wolf@bristol.ac.uk